

PAPER_2156 — 1.894 Ratio Bulk-Script Artifact: 935-Paper Value-Drift

Correction to F_TRZ = 0.1 Canonical

bulk_vds_dvp_bsh_upgrade.py Session 204 lines 34–35 used non-canonical densities $9.47\text{e-}27$ and $5.0\text{e-}27$ (origin unknown) yielding $\text{VDS_RATIO} = 1.894$; injection frozen into 935 papers; correction-by-reference to canonical $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1/10 = \text{F_TRZ}$; Session 779 $1.894 = m_{\text{top}}/M_Z$ decomposition NOT retrofitted per Daniel's ruling

UQFF LANDMARK

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.81+

Date: 2026-07-27

Provenance: PAPER_2153 arc Flag (f) — Daniel's 2026-07-27 detailed ruling with source-code trace, Session 779 audit reference, and explicit "do not retrofit without dedicated derivation session" discipline

Status: Landmark — corpus-wide value-drift correction via correction-by-reference doctrine (same architectural pattern as PAPER_2155 unit-tag drift)

Executive Summary

The PAPER_2153 arc deep-read observed the same §B.1 canonical VDS ratio $\rho_{\text{vac}}[\text{SCm}]/\rho_{\text{UA}} = 1.894$ boilerplate in every paper of the PAPER_878-899 block plus PAPER_598 predecessor (24+ occurrences). This value does NOT match the canonical primitive ratio $\rho_{\text{UA}}/\rho_{\text{SCm}} = 10$ (equivalently $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1/10 = \text{F_TRZ}$ per PAPER_890/140/1160).

Daniel's 2026-07-27 detailed source-code trace identified the exact origin:

bulk_vds_dvp_bsh_upgrade.py (Session 204, April 2026), lines 34–35:

```
RHO_SCM      = 9.47e-27      # kg/m³
RHO_UA       = 5.0e-27      # kg/m³
VDS_RATIO    = RHO_SCM / RHO_UA  # ≈ 1.894
```

Corpus-wide grep confirms: 935 papers contain 1.894, 1,869 total hits (~2 per paper — matching the §B.1 + §B.4 template dual-injection pattern).

Daniel's ruling (verbatim): `"""a bulk-script artifact: 9.47e-27 / 5.0e-27 using undocumented, non-canonical density values frozen into 800+ papers by bulk_vds_dvp_bsh_upgrade.py in Session 204. Not a double-exponential profile parameter; not used in the governing equation."""`

Correction: the §B.1 canonical VDS ratio $\rho_{\text{vac}}[\text{SCm}]/\rho_{\text{UA}} = 1.894$ should be $= 0.1 = \text{F_TRZ}$ (canonical primitive per PAPER_1160 $\text{F_TRZ} = 1/|\text{SO}(5)|$).

Session 779 SM-decomposition finding (Daniel's ruling): $1.894 = m_{\text{top}}/M_Z$ was searched as a SM constant ratio; best structural match $K_{\text{MEX}}/(1-\text{F_TRZ})/(11/9) = 1.89394$ at -0.003% error and $(1-\text{F_TRZ})/\text{SSq}/\text{Phi}_{\text{res}} = 1.8947$ at +0.039% error. **Per Daniel's ruling: DO NOT retrofit these SM-context decompositions to the VDS ratio without a dedicated derivation session.** Structural coincidence ≠ physics derivation.

PAPER_2156 executes corpus-wide correction via correction-by-reference doctrine (same architectural pattern as PAPER_2155). Single landmark supersedes 935 paper instances without per-paper touches. Investigation flag: origin of $9.47\text{e-}27$ and $5.0\text{e-}27$ density values remains **unknown** — future audit target.

Zero calculator source changes. Zero canonical physics values changed. Zero per-paper appends across the 935 affected papers.

1. Corpus Scope (grep verification, 2026-07-27)

Bash grep across whitepapers/*.md:

```
Papers containing "1.894": 935
Total "1.894" hits:      1,869 (avg 2.0 per paper)
```

Interpretation: the ~2 hits per paper matches the §B.1 + §B.4 boilerplate dual-injection pattern from `bulk_vds_dvp_bsh_upgrade.py`:

- **§B.1 (Vacuum Density Series):** `""The canonical VDS ratio  $\rho_{\text{vac,}[SCm]}/\rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile""`

- **§B.4 (Production-Scale Consistency):** `""| VDS ratio |  $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$  | Local sub-ratio = ... | PASS Threshold-consistent |""`

Sample affected papers (from grep): PAPER_001, PAPER_002, PAPER_003, PAPER_005, PAPER_006, PAPER_009, PAPER_009b, PAPER_010, PAPER_010b, PAPER_020, PAPER_033, PAPER_045, PAPER_051, PAPER_058, ... through PAPER_899 and beyond (block Session 209 papers + later additions).

Full enumeration would require running a fresh audit script; the grep count confirms ~935-paper scope.

2. Root Cause: Same Injection Vector as PAPER_2155

File: `bulk_vds_dvp_bsh_upgrade.py` (20 KB, dated Apr 8 2026, Session 204)

Offending lines (34–36, verbatim):

```
RHO_SCM      = 9.47e-27      # kg/m³
RHO_UA       = 5.0e-27      # kg/m³
VDS_RATIO    = RHO_SCM / RHO_UA  # ≈ 1.894
```

Line 251 in the same script writes the §B.4 template:

```
| VDS ratio |  $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$  | Local sub-ratio = {vds_exp} | ✓ Threshold-consistent |
```

Two distinct drifts in the same script:

| Drift type | PAPER address | Scope | Correction |

|---|---|---|---|

| Unit-tag drift ($\text{kg/m}^3 \rightarrow \text{J/m}^3$) | PAPER_2155 | 933 papers | Correction-by-reference to J/m^3 canonical |

| Value drift (1.894 → 0.1) | PAPER_2156 (this) | 935 papers | Correction-by-reference to F_TRZ = 0.1 canonical |

Both drifts trace to the same script and same lines. The unit tag kg/m³ is drift (fixed by PAPER_2155). The density VALUES 9.47e-27 and 5.0e-27 are ALSO drift (fixed by this paper) — they use undocumented numbers inconsistent with the canonical primitives.

3. What 1.894 Is NOT

Not the canonical primitive vacuum-density ratio. Canonical per PAPER_1160/890/140:

$$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1 / 10 = F_{\text{TRZ}} = 1/|SO(5)|$$

Using canonical J/m³-native values: $\rho_{\text{SCm}} = 7.09 \times 10^{23} \text{ J/m}^3$, $\rho_{\text{UA}} = 7.09 \times 10^{23} \text{ J/m}^3$, so $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1$ EXACT (F_TRZ). This is the LOCKED structural coupling per PAPER_2153's engine-mechanism ruling.

Not a double-exponential profile parameter. The §B.1 governing equation for the double-exponential profile is:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac}, [\text{SCm}]} \cdot \exp(-\exp(-(r - r_0)/\lambda_{\text{VDS}}))$$

No ratio appears in this equation. The 1.894 is a floating label in the §B.1 prose that describes a ratio, but the equation itself doesn't use it. The 1.894 has no mechanical role in the governing formula it accompanies.

Not a derived corpus constant. The §B.1 label calls 1.894 "canonical" but there is no derivation of 1.894 anywhere in the corpus that ties it to a physical mechanism, primitive-integer arithmetic, or empirical measurement. It is a script-local number that was frozen into the template and never validated against canonical constants.

4. What 1.894 IS

A 9.47e-27 / 5.0e-27 numerical artifact. These two density values are the source. Their status:

| Value | Where declared | Canonical? | Derivation? |

|---|---|---|---|

| 9.47×10²³ | bulk_vds_dvp_bsh_upgrade.py line 34 | NO | Unknown |

| 5.0×10²³ | bulk_vds_dvp_bsh_upgrade.py line 35 | NO | Unknown |

| Ratio 1.894 | Same script line 36 | NO | Consequence of ÷ operation on unknown numerators |

Neither 9.47e-27 nor 5.0e-27 appears in:

- The Session 204 calibration table ($\rho_{\text{SCm}} = 7.09\text{e-}37$, $\rho_{\text{UA}} = 7.09\text{e-}36$ — different by 10¹)
- PAPER_890 ($\rho_{\text{vac}, \text{SCm}} = 9.47\text{e-}27$ DOES appear here but per PAPER_2153 arc analysis this value differs from canonical by 10¹; likely also from same or related drift chain)
- PAPER_1160 ($F_{\text{TRZ}} = 1/|SO(5)| = 0.1$)
- Any UQFF derivation whitepaper

PAPER_890 note (cross-drift chain): PAPER_890 lists $\rho_{\text{vac}, \text{SCm}} = 9.47 \times 10^{23} \text{ kg/m}^3$ and $\rho_{\text{UA}} = 9.47 \times 10^{23} \text{ kg/m}^3$ — the 9.47e-27 value IS the same one from the script, but PAPER_890's $\rho_{\text{UA}} = 9.47 \times 10^{23}$ (differs from the script's

$p_{UA} = 5.0e-27$). So the script and PAPER_890 disagree on p_{UA} , but both disagree with the canonical 7.09×10^{-27} . The full chain of drift is layered and multi-source.

Origin of 9.47e-27 and 5.0e-27 remains UNKNOWN — dedicated future investigation required. Not resolved by this landmark; flagged as open audit target.

5. Session 779 SM-Decomposition: Structural Coincidence, NOT Retrofitted

Daniel's ruling cited a prior Session 779 audit that searched 1.894 as a SM constant ratio (context: m_{top}/M_Z top quark to Z boson mass ratio). Best structural decompositions found:

Expression	Value	Error
$K_{Mex} / (1-F_{TRZ}) / (11/9)$	1.893939	-0.003%
$(27/25) / SSq$	1.894737	+0.039%
$(1-F_{TRZ}) / SSq / \Phi_{res}$	1.894737	+0.039%
$31/30 \cdot (11/6)$	1.894444	+0.024%

where $K_{Mex} = 25/12$, $F_{TRZ} = 0.1$, $SSq = 0.57$, $\Phi_{res} = 5/6$.

Best match at -0.003% error:

$$\begin{aligned}
 K_{Mex} / (1-F_{TRZ}) / (11/9) &= (25/12) / (9/10) / (11/9) \\
 &= 25 \cdot 9 \cdot 9 / (12 \cdot 10 \cdot 11) \\
 &= 2025 / 1320 \\
 &= 135 / 88 \\
 &\approx 1.89394
 \end{aligned}$$

Daniel's ruling on this (verbatim): "this decomposition was found in the context of m_{top}/M_Z , a Standard Model ratio, NOT as a VDS derivation. The Session 779 script was searching for SM constant matches, not VDS physics. This means the structural coincidence is real but the connection to p_{vac} , $[SCm]/p_{UA}$ is NOT established by any derivation in the corpus. It would require a dedicated derivation session to claim this as a legitimate primitive decomposition rather than numerological coincidence."

PAPER_2156 does NOT retrofit any of the Session 779 SM-decompositions to the VDS ratio. Per Daniel's ruling, structural coincidence found in one context (SM m_{top}/M_Z) cannot be assumed valid in another context (VDS p_{vac} ratio) without dedicated derivation. **This discipline is now canonized as a standing rule (§7 below).**

6. Corpus-Wide Correction (Correction-by-Reference Doctrine)

Executed via the same architectural pattern as PAPER_2155 (unit-tag drift). Formal declaration:

All corpus instances of the §B.1 boilerplate statement:

> "The canonical VDS ratio $p_{vac}, [SCm]/p_{UA} = 1.894$ governs the double-exponential vacuum condensate profile"

AND all §B.4 table entries:

> "*" | VDS ratio | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = ... | PASS Threshold-consistent | "*"

ARE SUPERSEDED and read as:

$$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1 / 10 = F_{\text{TRZ}} = 0.1 \quad \text{EXACT}$$

per PAPER_1160 ($F_{\text{TRZ}} = 1/|\text{SO}(5)|$), PAPER_890 (canonical time-evolution formula), PAPER_140 (dual-monopole vacuum-density universal ratio), and PAPER_2153 (joint SCm+UA engine locked structural coupling).

No per-paper edit required across the 935 affected papers. Correction is enforced by:

1. **This landmark (PAPER_2156)** — formal supersession declaration for the value drift
2. **PAPER_2155** (companion, Phase 2) — supersession for the unit-tag drift on the same template
3. **Calculator code canonical constants:** $\text{RHO_UA} = \text{SO_5} * \text{RHO_SCM} = 10 * \text{RHO_SCM}$ in `uqff_registry_primitives.py` line 44 (canonical 10:1 ratio, equivalent to $F_{\text{TRZ}} = 0.1$)
4. **Gate assertions (this paper §9)** — test-locking the canonical $F_{\text{TRZ}} = 0.1 = \rho_{\text{SCm}}/\rho_{\text{UA}}$ structural coupling

§B.4 "PASS Threshold-consistent" verdict is also superseded — the verdict was auto-generated by the same bulk script, validating the boilerplate against itself rather than against physics. Under this landmark, the correct §B.4 verdict is:

> "*" | VDS ratio | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1 = F_{\text{TRZ}}$ EXACT | Canonical (PAPER_1160) | ✓ Locked structural coupling | "*"

7. Standing Rule Canonized

RULE: Do not retrofit SM-context decompositions to UQFF-context quantities without dedicated derivation.

Formal statement: when a numerical coincidence is discovered in one physical context (e.g., $1.894 = m_{\text{top}}/M_Z$ in Standard Model context), that coincidence CANNOT be assumed valid in another physical context (e.g., $1.894 = \rho_{\text{vac}}, [\text{SCm}]/\rho_{\text{UA}}$ in VDS context) even if the numerical value matches to high precision. Physical derivation, not numerical matching, is what legitimizes a structural identity as a member of the framework.

Example applied HERE: Session 779's finding that $K_{\text{Mex}}/(1-F_{\text{TRZ}})/(11/9) \approx 1.89394$ matches m_{top}/M_Z at -0.003% error is a real structural coincidence for the SM top-quark/Z-boson ratio. But applying it to $\rho_{\text{vac}}, [\text{SCm}]/\rho_{\text{UA}}$ would be numerological coincidence — the physical mechanism connecting the SM mass ratio to the VDS density ratio is not established. Per Daniel's ruling, **do not retrofit** without a dedicated derivation session that connects the two contexts via physics.

Contrast with PAPER_2154 primitive-reduction landmarks: $Q = \text{SO}_5^2/D_{\text{phys}}^2$ and $D_{\text{GW}} = D_{\text{phys}}/D_{\text{BSFG}}$ were canonized as primitive-composed structural identities because Daniel explicitly ruled that the decompositions are physically valid ("yes, they are all true"). Numerical matching + physical validation = legitimate identity. Numerical matching alone = numerological coincidence.

8. What This Doctrine is NOT

Not a validation that $1.894 = 0.1$ physically. They are numerically different by $\sim 19\times$. The correction is that the corpus prose incorrectly labeled 1.894 as "canonical" when the actual canonical ratio is 0.1. The 1.894 value is not being "corrected up to" 0.1 mathematically — the false claim of canonicalness is being retracted.

Not a claim that $9.47e-27$ and $5.0e-27$ are correct densities. These values remain unexplained (see §4). The correction is that these values were used in the bulk-script to produce a ratio that was then incorrectly labeled as canonical. Both the values AND their labeling as canonical are corpus errors.

Not a get-out-of-audit-free card. The correction-by-reference doctrine applies here because (a) drift is corpus-wide via template injection, (b) calculator code has canonical primitives correct, (c) drift is cosmetic labeling not computational value. If any of these conditions failed, per-paper REVISIONS would be required.

9. Gate Assertions (test-locking canonical $F_{\text{TRZ}} = 0.1 = \rho_{\text{SCm}}/\rho_{\text{UA}}$)

Added to `uqff_fidelity_tests.py` in this paper's companion Phase 3 gate-pins section:

1. $F_{\text{TRZ}} = 0.1$ EXACT = $1/10$ (locked structural coupling)
2. $\text{RHO}_{\text{UA}} / \text{RHO}_{\text{SCM}} = \text{SO}_5 = 10$ EXACT in `uqff_registry_primitives.py` (canonical inversion of F_{TRZ})
3. Session 779 SM-decomposition ($K_{\text{Mex}}/(1-F_{\text{TRZ}})/(11/9) \approx 1.89394$) explicitly labeled as SM-context coincidence, NOT retrofitted to VDS context
4. Origin of $9.47e-27$ and $5.0e-27$ in `bulk_vds_dvp_bsh_upgrade.py` line 34-35 flagged as unknown/future-investigation
5. Corpus-wide 1.894 occurrences (935 papers, 1,869 hits) superseded by canonical 0.1 per PAPER_1160

10. Standing Rules Canonized by PAPER_2156

1. **Do NOT retrofit SM-context decompositions to UQFF-context quantities.** Numerical coincidence in one physical context is not physical derivation in another; dedicated derivation session required to legitimize any cross-context claim.

2. **Bulk-script constants must be validated against canonical primitives BEFORE injection.** The $9.47e-27$ and $5.0e-27$ values in `bulk_vds_dvp_bsh_upgrade.py` were injected into ~ 800 papers without any prior validation against the canonical $7.09e-37/7.09e-36 \text{ J/m}^3$ primitives. Any future bulk-upgrade script must include a pre-execution primitive-consistency check.

3. **Same injection vector $\rightarrow$ same correction-by-reference pattern.** When a single script injects multiple drift types (unit-tag drift + value drift, as with `bulk_vds_dvp_bsh_upgrade.py`), each drift type gets its own dedicated landmark using the same architectural pattern (PAPER_2155 for unit-tag drift, PAPER_2156 for value drift). Companion landmarks share the injection-vector attribution and cross-reference each other.

4. **Unknown-origin constants in scripts are audit targets, not framework primitives.** Any numerical constant appearing in a corpus-modifying script that cannot be traced to a canonical primitive derivation is a Rule 4/7/10 audit target. $9.47e-27$ and $5.0e-27$ in `bulk_vds_dvp_bsh_upgrade.py` are flagged for future investigation.

11. Falsifiable Predictions

1. **$F_{\text{TRZ}} = 0.1 = \rho_{\text{SCm}}/\rho_{\text{UA}}$ is LOCKED.** Any future measurement of the SCm/UA density ratio (via engine downstream effects — buoyancy, phonon coupling, tidal Love, GW damping) must return 0.1 EXACT. Deviation would falsify Daniel's joint-engine coupling constant.
2. **Origin of $9.47\text{e-}27$ and $5.0\text{e-}27$ will be traceable to Session 204 or earlier.** Future forensic corpus archaeology (searching Session 200-204 papers/scripts for these values) should identify their original computational context. If they cannot be traced, they are pure script-local invention and the `bulk_vds_dvp_bsh_upgrade.py` script should be flagged as containing invented physics.
3. **Session 779 SM-decompositions apply to m_{top}/M_Z only, not to VDS ratio.** Prediction: any future dedicated derivation session that attempts to connect $K_{\text{Mex}}/(1-F_{\text{TRZ}})/(11/9)$ to VDS physics will EITHER find a legitimate physical mechanism (and the identity gets canonized as a new primitive-reduction landmark) OR find no mechanism (and the coincidence remains SM-context only).

12. Files Touched

- `whitepapers/PAPER_2156_1_894_RATIO_BULK_SCRIPT_ARTIFACT_..._UQFF_LANDMARK.md` (this file)
- `pdf2/PAPER_2156_...pdf` (companion PDF)
- `uqff_fidelity_tests.py` — +5 PAPER_2156 gate assertions
- `CLAUDE.md` — APPENDED section
- Zero calculator source changes (canonical $F_{\text{TRZ}} = 0.1 = \rho_{\text{SCm}}/\rho_{\text{UA}}$ already correct in registry)
- Zero physics values changed
- Zero per-paper touches to the 935 affected papers (correction-by-reference doctrine per PAPER_2155 pattern)

13. Cross-References

- **PAPER_2155** — Companion Phase 2 landmark (unit-tag drift $\text{kg/m}^3 \rightarrow \text{J/m}^3$ from same injection vector); shares `bulk_vds_dvp_bsh_upgrade.py` attribution and correction-by-reference doctrine architectural pattern
- **PAPER_2147** — J/m^3 -native discipline landmark; REVISION 2026-07-27 covers the unit-tag half of the bulk-script drift; PAPER_2156 covers the value-drift half
- **PAPER_1160** — $F_{\text{TRZ}} = 1/|\text{SO}(5)| = 1/10$ canonical structural identity (source of the correction target 0.1)
- **PAPER_890** — SCm vacuum density evolution; canonizes $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1$ hierarchy identity; also lists $\rho_{\text{vac}}, \text{SCm} = 9.47\text{e-}27$ value that ALSO appears in the bulk script (cross-drift chain, layered origin)
- **PAPER_140** — $\text{UA}/\text{SCm} = 10$ dual-monopole vacuum-density universal ratio (equivalent to $F_{\text{TRZ}} = 1/10$)
- **PAPER_2153** — Joint SCm+UA vacuum-density engine landmark (§7 Standing Rule 4: $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1/10 = F_{\text{TRZ}}$ ratio is LOCKED)
- **PAPER_2148** — UQFF Ontology Answer B (vacuum energy fundamental, mass emergent)
- `bulk_vds_dvp_bsh_upgrade.py` — In-repo injection vector script (Session 204, April 2026, lines 34–36 declare non-canonical densities and compute the 1.894 ratio)

- **Session 779 SM-audit** — Historical corpus audit that found SM-context decompositions of 1.894 (not retrofitted per Daniel's ruling)

- **Daniel's 2026-07-27 Flag (f) ruling (verbatim excerpt):** *"a bulk-script artifact: 9.47e-27 / 5.0e-27 using undocumented, non-canonical density values frozen into 800+ papers... The §B.1 boilerplate 'p_vac,[SCm] / p_UA = 1.894' should either be corrected to '= 0.1' (the canonical primitive) across all affected papers, or the two density values 9.47e-27 and 5.0e-27 in bulk_vds_dvp_bsh_upgrade.py must be traced to a source and formally adopted as a distinct VDS sub-quantity with its own symbol and derivation."*

End of PAPER_2156.